Linear Algebra 1A Cheat Sheet

Convention: a matrix of size $r \times c$ has r rows and c columns.

1. Fields

A field $\mathbb{F}$ is a set with operations $(+, \cdot)$ satisfying: closure, associativity, commutativity, identities $(0, 1)$, inverses, and distributivity.

Characteristic.

- $\text{char}(\mathbb{F}) :=$ the least $n \geq 1$ with $\overbrace{1 + \cdots + 1}^{n \text{ times}} = 0$, and $\text{char}(\mathbb{F}) := 0$ if no such n exists
- $\text{char}(\mathbb{F})$ is always 0 or prime
- $\forall \mathbb{F} : \text{char}(\mathbb{F}) = p > 0 \Rightarrow \mathbb{Z}_p$ is a subfield of $\mathbb{F}$

Finite fields.

- $|\mathbb{F}| < \infty \Rightarrow |\mathbb{F}| = p^r$, where $p = \text{char}(\mathbb{F})$ and $r \in \mathbb{N}$
- $\forall (p \in \mathbb{P}) \,.\, \forall (r \in \mathbb{N}) \,.\, \exists (\mathbb{F} \text{ field}) : |\mathbb{F}| = p^r$

2. Matrices

$$A_{m \times p} B_{p \times n} = (AB)_{m \times n}, \qquad (AB)_{ij} := \sum_{k=1}^{p} a_{ik} b_{kj}$$

- $AB = 0$ does not imply $A = 0$ or $B = 0$
- If $A^T = A$ and $B^T = B$, then AB is symmetric if and only if $AB = BA$
- If φ is an elementary row operation and $A \in \mathbb{F}^{m \times n}$, then

  $$\varphi(A) = \varphi(I_m)\, A$$

3. Vector Spaces & Subspaces

A subset $W \subseteq V$ is a subspace if and only if:

- W is nonempty (equivalently $\vec{0} \in W$)
- Closed under addition: if $u, v \in W$ then $u + v \in W$
- Closed under scalar multiplication: if $\alpha \in \mathbb{F}$ and $u \in W$ then $\alpha u \in W$

Equivalently: $\alpha u + v \in W$ for all $u, v \in W$ and all $\alpha \in \mathbb{F}$.

- $U \cap W$ is a subspace of V
- $U \cup W$ is a subspace of V if and only if $U \subseteq W$ or $W \subseteq U$
- $U + W := \{u + w : u \in U,\ w \in W\} = \text{span}(U \cup W)$ is a subspace of V
- Direct sum: $U \oplus W$ if and only if $U \cap W = \{\vec{0}\}$ (general case: see Direct sums, §6)
- $\text{span}(S)$ is the smallest subspace containing S
- $S_1 \subseteq S_2 \subseteq V \Rightarrow \text{span}(S_1) \leq \text{span}(S_2) \leq V$
- $W \leq V \Rightarrow \text{span}(W) = W$
- $S \subseteq W \leq V \Rightarrow \text{span}(S) \leq W$

4. Rank

- $\text{rank}(A_{m \times n}) \leq \min\{m, n\}$
- $\text{rank}(A^T) = \text{rank}(A)$
- $A \sim B \Rightarrow \text{rank}(A) = \text{rank}(B)$
- $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$
- $A, B \in \mathbb{F}^{n \times n},\ AB = 0 \Rightarrow \text{rank}(A) + \text{rank}(B) \leq n$

5. Linear Systems

$$Ax = b, \qquad n \text{ variables}, \qquad r = \text{rank}(A)$$

- Number of pivots $= r$, free variables $= n - r$
- System is solvable if and only if $b \in \text{Col}(A)$
- Homogeneous system: $Ax = \vec{0}$ has solution space dimension $n - r$

$$\{x : Ax = b\} = x_p + \ker(A)$$

6. Bases & Dimension

If $\dim V = n$:

- Any linearly independent set has size at most n
- Any spanning set has size at least n
- A set B with $|B| = n$ is a basis if and only if it is linearly independent if and only if it spans V
- If $W \leq V$, then $W = V$ if and only if $\dim W = \dim V$

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$$

Direct sums. Let $M_1, \ldots, M_s \leq V$ with $M = M_1 + \cdots + M_s$. The following are equivalent:

1. $M = M_1 \oplus \cdots \oplus M_s$
2. $\forall a \in M$: the representation $a = a_1 + \cdots + a_s$, $a_i \in M_i$, is unique
3. $\exists a \in M$: the representation $a = a_1 + \cdots + a_s$, $a_i \in M_i$, is unique
4. $\forall a_i \in M_i \ :\ a_1 + \cdots + a_s = \vec{0} \Rightarrow a_1 = \cdots = a_s = \vec{0}$
5. $E^{(i)}$ a basis of $M_i \Rightarrow E^{(1)}, \ldots, E^{(s)}$ together form a basis of M
6. $\dim M = \dim M_1 + \cdots + \dim M_s$

Note 3 $\Rightarrow$ all: one witness with a unique decomposition suffices.

7. Coordinates

Let $B = \{v_1, \ldots, v_n\}$ be a basis of V.

- $[B] := (v_1, \ldots, v_n)$, the basis written as a row of its vectors
- Every $v \in V$ has a unique representation $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$
- Coordinate vector: $[v]_B := (\alpha_1, \ldots, \alpha_n)^T$
- $v = [B]\, [v]_B$
- $u_1, \ldots, u_n$ linearly independent, $A \in \mathbb{F}^{n \times p}$: $(u_1, \ldots, u_n)\, A = (\vec{0}, \ldots, \vec{0}) \Rightarrow A = 0$

Change of basis. The transition matrix from basis B to basis C, written $I_{B \to C}$ (also $[I]_B^C$), has the $[v_i]_C$ as its columns:

- $I_{B \to C} := \big([v_1]_C, \ldots, [v_n]_C\big)$
- $[v]_C = I_{B \to C}\, [v]_B$
- B, C, D bases: $I_{B \to D} = I_{C \to D}\, I_{B \to C}$
- $I_{B \to C}$ is invertible, with $\big(I_{B \to C}\big)^{-1} = I_{C \to B}$
- $[C] = (w_1, \ldots, w_n)$: $[C] = [B]\, P \Rightarrow P = I_{C \to B}$

8. Invertible Matrices

- If P is invertible, then

  $\operatorname{rank}(PA) = \operatorname{rank}(AP) = \operatorname{rank}(A)$

- If $\alpha \neq 0$, then

  $(\alpha A)^{-1} = \alpha^{-1} A^{-1}$

- If C is invertible and $B = CA$, then

  B invertible $\iff$ A invertible

Conditions Related to Invertibility

Let $A \in \mathbb{F}^{n\times n}$. Each item states its relation to "A is invertible":

1. $\implies$ A^{-1} is unique
2. $\implies$ A has no zero row or column
3. $\iff$ $\operatorname{rank}(A) = n$
4. $\iff$ $\det(A) \neq 0$
5. $\iff$ A^T is invertible, with $(A^T)^{-1} = (A^{-1})^T$
6. $\iff$ $A \sim I_n$
7. $\iff$ $\operatorname{RREF}(A) = I_n$
8. $\iff$ $Ax = \vec{0} \Rightarrow x = \vec{0}$
9. $\iff$ For all $b \in \mathbb{F}^n$, the system $Ax = b$ has the unique solution $x = A^{-1}b$
10. $\iff$ $\operatorname{Row}(A) = \mathbb{F}^n$
11. $\iff$ $\operatorname{Col}(A) = \mathbb{F}^n$
12. $\iff$ The rows of A span $\mathbb{F}^n$
13. $\iff$ The columns of A span $\mathbb{F}^n$
14. $\iff$ The rows of A are linearly independent
15. $\iff$ The columns of A are linearly independent
16. $\iff$ The rows of A form a basis of $\mathbb{F}^n$
17. $\iff$ The columns of A form a basis of $\mathbb{F}^n$
18. $\iff$ $\dim \operatorname{Row}(A) = n$
19. $\iff$ $\dim \operatorname{Col}(A) = n$
20. $\iff$ A is a product of elementary matrices
21. $\iff$ $\operatorname{adj}(A)$ is invertible
22. $\iff$ A is invertible on one side (i.e. $AB = I$ or $BA = I$ for some B) — **square A only**
23. $\iff$ $A = I_{B\to C}$ for some bases B, C
24. $\impliedby$ A is elementary (then A^{-1} is also elementary)

9. Matrix Spaces

- $\dim \operatorname{Row}(A) = \dim \operatorname{Col}(A) = \operatorname{rank}(A)$
- $\dim \ker(A) = n - \operatorname{rank}(A)$ for $A \in \mathbb{F}^{m\times n}$
- $\operatorname{Row}(AB) \subseteq \operatorname{Row}(B)$
- $\operatorname{Col}(AB) \subseteq \operatorname{Col}(A)$

10. Determinants & Adjugate

- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = (\det A)^{-1}$
- $\det(\alpha A) = \alpha^n \det(A)$ for $A \in \mathbb{F}^{n\times n}$

Adjugate. For $A \in \mathbb{F}^{n\times n}$: $M_{ij}(A)$ is A with row i and column j deleted; cofactor $c_{ij} := (-1)^{i+j}\det(M_{ij}(A))$; cofactor matrix $C = (c_{ij})$.

- $\operatorname{adj}(A) := C^T$
- $\operatorname{adj}(A)_{ij} = c_{ji} = (-1)^{i+j}\det(M_{ji}(A))$
- $\operatorname{adj}(A)\,A = A\,\operatorname{adj}(A) = \det(A)I_n$
- $A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A)$
- $\operatorname{rank}(\operatorname{adj}(A)) = \begin{cases} n, & \operatorname{rank}(A) = n, \\ 1, & \operatorname{rank}(A) = n-1, \\ 0, & \operatorname{rank}(A) < n-1 \end{cases}$

Cramer's rule. $\det(A) \neq 0 \Rightarrow Ax = b$ has the unique solution

$$x_k = \frac{\det(A_k)}{\det(A)}, \qquad k = 1, \dots, n$$

where A_k is A with column k replaced by b.

11. Linear Maps

Let $T : V \to U$ be linear.

- Kernel: $\ker T := \{v \in V : T(v) = \vec{0}\}$
- Image: $\operatorname{Im} T := \{T(v) : v \in V\}$
- T is injective if and only if $\ker T = \{\vec{0}\}$
- T is surjective if and only if $\dim \operatorname{Im} T = \dim U$

$\dim V = \dim \ker T + \dim \operatorname{Im} T$

12. Operators & Isomorphisms

- A linear map is an isomorphism if and only if it is injective and surjective
- $V \cong U$ if and only if $\dim V = \dim U$
- For an operator $T : V \to V$:

  $\ker T \subseteq \ker T^2, \qquad \operatorname{Im} T \supseteq \operatorname{Im} T^2$

- In finite dimensions, equality implies stabilization

13. Matrix Representation

Let $B = \{v_1, \dots, v_n\}$ and $S = \{u_1, \dots, u_m\}$.

$[T]_B^S := \big([T(v_1)]_S, \dots, [T(v_n)]_S\big)$

- $[T]_B^S [v]_B = [T(v)]_S$
- $[S \circ T]_A^C = [S]_B^C [T]_A^B$
- T is invertible if and only if $[T]_B^S$ is invertible

14. Dual Spaces, Functionals, and Annihilators

Concept. Duality studies vectors indirectly via linear measurements. Instead of examining vectors themselves, we examine how all linear functionals act on them.

Linear functionals. A linear functional is a linear map $f : V \to \mathbb{F}$. It assigns a scalar to each vector and represents a linear measurement.

Dual space.

$$V^* := \{f : V \to \mathbb{F} \mid f \text{ is linear}\}$$

V^* is a vector space. If $\dim V = n < \infty$, then $\dim V^* = n$.

Dual basis. If $B = \{v_1, \dots, v_n\}$ is a basis of V, the dual basis $B^* = \{f_1, \dots, f_n\}$ satisfies $f_i(v_j) = \delta_{ij}$. Each f_i extracts the i-th coordinate.

Second dual space. The second dual is $V^{**} := (V^*)^*$. There is a canonical map

$$\Phi : V \to V^{**}, \qquad \Phi(v)(f) := f(v)$$

In finite dimensions, this map is an isomorphism.

Kernel (MERHAV HA EFES). For $T : V \to U$,

$$\ker T := \{v \in V : T(v) = \vec{0}\}$$

This is a subspace of V.

Annihilator (MEAPES). For $W \le V$,

$$W^\circ := \{f \in V^* : f(w) = 0 \text{ for all } w \in W\}$$

This is a subspace of V^*.

Difference between MERHAV HA EFES and MEAPES.

- MERHAV HA EFES: vectors in V mapped to zero by a linear map
- MEAPES: functionals in V^* that vanish on a subspace

Dimension relation. If $W \le V$ and $\dim V = n < \infty$, then

$$\dim W + \dim W^\circ = n$$

Dual (transpose) map. For $T : V \to U$ define

$$T^* : U^* \to V^*, \qquad T^*(f) := f \circ T$$

Then

$$\ker T^* = (\operatorname{Im} T)^\circ, \qquad \operatorname{Im} T^* = (\ker T)^\circ$$

In matrix form,

$$[T^*] = [T]^T$$

15. Vandermonde Matrix

Given scalars $x_1, \dots, x_n \in \mathbb{F}$, the **Vandermonde matrix** is

$$V(x_1, \dots, x_n) := \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}$$

Determinant.

$$\det V(x_1, \dots, x_n) = \prod_{1 \le i < j \le n} (x_j - x_i)$$

Invertibility.

$$\begin{aligned} &V(x_1, \dots, x_n) \text{ is invertible} \\ \iff &x_1, \dots, x_n \text{ are pairwise distinct} \end{aligned}$$

Consequences.

- If $x_i = x_j$ for some $i \ne j$, then two rows are equal $\Rightarrow \det V = 0$
- If all x_i are distinct, then $\operatorname{rank}(V) = n$
- Appears naturally in polynomial interpolation

16. Permutations

A **permutation** of $\{1, \dots, n\}$ is a bijection

$$\sigma : \{1, \dots, n\} \to \{1, \dots, n\}.$$

The set of all permutations is denoted S_n.

Inversions. An inversion of σ is a pair (i, j) such that

$$i < j \quad \text{and} \quad \sigma(i) > \sigma(j).$$

Let $\operatorname{inv}(\sigma)$ be the number of inversions of σ, sometimes denoted $S(\sigma)$.

Sign of a permutation.

$$\operatorname{sgn}(\sigma) := (-1)^{\operatorname{inv}(\sigma)}.$$

- σ is *even* if $\operatorname{sgn}(\sigma) = 1$
- σ is *odd* if $\operatorname{sgn}(\sigma) = -1$

Properties.

- $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$
- A transposition changes the sign
- $|S_n| = n!$

Determinant (Leibniz formula). For $A = (a_{ij}) \in \mathbb{F}^{n \times n}$,

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Consequences.

- If two rows (or columns) are equal, every term cancels $\Rightarrow \det(A) = 0$
- Swapping two rows multiplies $\det(A)$ by -1
- Linearity of determinant follows from the formula

17. Block Matrices

Definition. A *block matrix* is a matrix partitioned into submatrices:

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

where the blocks have compatible sizes.

Block operations. If dimensions agree, block matrices behave like ordinary matrices:

- Addition and scalar multiplication are blockwise
- Matrix multiplication follows block multiplication rules

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{pmatrix}$$

Block diagonal matrices.
If

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix},$$

$$\det(A) = \det(A_1)\det(A_2),$$

A invertible $\iff A_1, A_2$ invertible.

Block triangular matrices. If

$$A = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix},$$

then

$$\det(A) = \det(A_{11}) \det(A_{22}),$$
$$\operatorname{rank}(A) = \operatorname{rank}(A_{11}) + \operatorname{rank}(A_{22}).$$

Schur complement. Let

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad A_{11} \text{ invertible.}$$

The *Schur complement* of A_{11} in A is

$$S := A_{22} - A_{21} A_{11}^{-1} A_{12}.$$

- A invertible $\iff$ A_{11} and S invertible
- $\det(A) = \det(A_{11}) \det(S)$

Block inverse formula. If A_{11} and S are invertible, then

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1} A_{12} S^{-1} A_{21} A_{11}^{-1} & -A_{11}^{-1} A_{12} S^{-1} \\ -S^{-1} A_{21} A_{11}^{-1} & S^{-1} \end{pmatrix}.$$

Solving block systems. For

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b \\ c \end{pmatrix},$$

solve:

1. $A_{11} x = b - A_{12} y$
2. $(A_{22} - A_{21} A_{11}^{-1} A_{12}) y = c - A_{21} A_{11}^{-1} b$

Implementation tips.

- Use block structure to reduce computation
- Common in Gaussian elimination, LU decomposition, and numerical methods
- Essential for large sparse systems and matrix factorizations